

Introduction to Artificial Intelligence

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AI is a technique that facilitates a machine to perform all cognitive functions such as perceiving, learning and reasoning that are otherwise performed by humans.

“The Science and Engineering of making intelligent machines, especially intelligent Computer programs is Artificial intelligence” –John McCarthy [Father of AI]

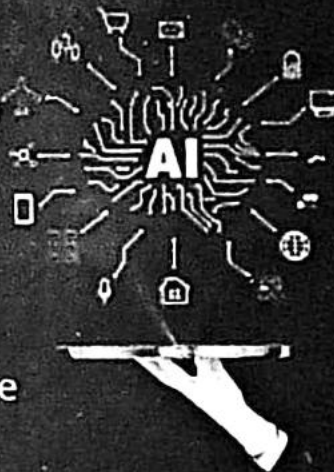
What Is AI or Artificial Intelligence?

Artificial intelligence or AI is the branch of computer science that studies machine intelligence.

EXAMPLES OF APPLICATIONS

- Search engines (Google)
- Content recommendations (Netflix, YouTube)
- Self-driving vehicles
- Automatic language translation
- Facial recognition
- Computer games
- Spam filters

An AI is a computer system that performs tasks that usually require human intelligence.



1.1 History & overview of Artificial Intelligence

- Introduction
- Overview of 'Artificial' Intelligence
- Themes of AI
- History of AI

• Introduction

Artificial Intelligence (AI) has been studied for decades and is still one of the most elusive subjects in Computer Science. AI ranges from machines truly capable of thinking to search algorithms used to play board games. It has applications in nearly every way we use computers in society.

• Overview of 'Artificial' Intelligence

The term artificial intelligence was first coined by John McCarthy in 1956 when he held the first academic conference on the subject.

But the journey to understand if machines can truly think began much before that. In Vannevar Bush's seminal work *As We May Think* [Bush45] he proposed a system which amplifies people's own knowledge and understanding.

Five years later Alan Turing wrote a paper on the notion of machines being able to simulate human beings and the ability to do intelligent things, such as play Chess [Turing50]. No one can refute a computer's ability to process logic. But to many it is unknown if a machine can think. The precise definition of think is important because there has been some strong they were passed notes in Chinese. Using an entire library of rules and look-up tables they would be able to produce valid responses in Chinese, but would they

really 'understand' the language? The argument is that since computers would always be applying rote fact lookup they could never 'understand' a subject. This argument has been refuted in numerous ways by researchers, but it does undermine people's faith in machines and so-called expert systems in life-critical applications.

• Themes of AI

The main advances over the past sixty years have been advances in search algorithms, machine learning algorithms, and integrating statistical analysis into understanding the world at large. However most of the breakthroughs in AI aren't noticeable to most people. Rather than talking machines used to pilot space ships to Jupiter, AI is used in more subtle ways such as examining purchase histories and influence marketing decisions [Shaw01].

What most people think of as 'true AI' hasn't experienced rapid progress over the decades. A common theme in the field has been to overestimate the difficulty of foundational problems. Significant AI breakthroughs have been promised 'in 10 years' for the past 60 years.

In addition, there is a tendency to redefine what 'intelligent' means after machines have mastered an area or problem. This so-called 'AI Effect' contributed to the downfall of US-based AI research in the 80s. In the field of AI expectations seem to always outpace the reality. After decades of research, no computer has come close to passing the Turing Test (a model for measuring 'intelligence'); Expert Systems have grown but have not become as common as human experts; and while we've built software that can beat humans at some games, open ended games are still

far from the mastery of computers. Is the problem simply that we haven't focused enough resources on basic research, as is seen in the AI winter section, or is the complexity of AI one that we haven't yet come to grasp yet? (And instead, like in the case of computer Chess, we focus on much more specialized problems rather than understanding the notion of 'understanding' in a problem domain.)

History of AI

1950

Alan Turing published "Computing Machinery and Intelligence," introducing the Turing test and opening the doors to what would be known as AI.

1951

Marvin Minsky and Dean Edmonds developed the first artificial neural network (ANN) called SNARC using 3,000 vacuum tubes to simulate a network of 40 neurons.

1952

Arthur Samuel developed Samuel Checkers-Playing Program, the world's first program to play games that was self-learning.

1956

John McCarthy, Marvin Minsky, Nathaniel Rochester and Claude Shannon coined the term artificial intelligence in a proposal for a workshop widely recognized as a founding event in the AI field.

1958

Frank Rosenblatt developed the perceptron, an early ANN that could learn from data and became the foundation for modern neural networks. John McCarthy developed the programming language Lisp, which was quickly adopted by the AI industry

and gained enormous popularity among developers.

1959

Arthur Samuel coined the term *machine learning* in a seminal paper explaining that the computer could be programmed to outplay its programmer Oliver Selfridge published "Pandemonium: A Paradigm for Learning," a landmark contribution to machine learning that described a model that could adaptively improve itself to find patterns in events.

1964

Daniel Bobrow developed STUDENT, an early natural language processing (NLP) program designed to solve algebra word problems, while he was a doctoral candidate at MIT.

1965

Edward Feigenbaum, Bruce G. Buchanan, Joshua Lederberg and Carl Djerassi developed the first expert system, Dendral, which assisted organic chemists in identifying unknown organic molecules.

1966

Joseph Weizenbaum created Eliza, one of the more celebrated computer programs of all time, capable of engaging in conversations with humans and making them believe the software had humanlike emotions.

Stanford Research Institute developed Shakey, the world's first mobile intelligent robot that combined AI, computer vision, navigation and NLP. It's the grandfather of self-driving cars and drones.

1968

Terry Winograd created SHRDLU, the first multimodal AI that could manipulate and reason out a world of blocks according to

instructions from a user.

1969

Arthur Bryson and Yu-Chi Ho described a backpropagation learning algorithm to enable multilayer ANNs, an advancement over the perceptron and a foundation for deep learning. Marvin Minsky and Seymour Papert published the book *Perceptrons*, which described the limitations of simple neural networks and caused neural network research to decline and symbolic AI research to thrive.

1973

James Lighthill released the report "Artificial Intelligence: A General Survey," which caused the British government to significantly reduce support for AI research.

1980

Symbolics Lisp machines were commercialized, signaling an AI renaissance. Years later, the Lisp machine market collapsed.

1981

Danny Hillis designed parallel computers for AI and other computational tasks, an architecture similar to modern GPUs.

1984

Marvin Minsky and Roger Schank coined the term AI winter at a meeting of the Association for the Advancement of Artificial Intelligence, warning the business community that AI hype would lead to disappointment and the collapse of the industry, which happened three years later.

1985

Judea Pearl introduced Bayesian networks causal analysis, which provides statistical techniques for representing uncertainty in computers.

1988

Peter Brown et al. published "A Statistical Approach to Language Translation," paving the way for one of the more widely studied machine translation methods.

1989

Yann LeCun, Yoshua Bengio and Patrick Haffner demonstrated how convolutional neural networks (CNNs) can be used to recognize handwritten characters, showing that neural networks could be applied to real-world problems.

1997

Sepp Hochreiter and Jürgen Schmidhuber proposed the Long Short-Term Memory recurrent neural network, which could process entire sequences of data such as speech or video.

IBM's Deep Blue defeated Garry Kasparov in a historic chess rematch, the first defeat of a reigning world chess champion by a computer under tournament conditions.

2000

University of Montreal researchers published "A Neural Probabilistic Language Model," which suggested a method to model language using feedforward neural networks.

2006

Fei-Fei Li started working on the ImageNet visual database, introduced in 2009, which became a catalyst for the AI boom and the basis of an annual competition for image recognition algorithms.

IBM Watson originated with the initial goal of beating a human on the iconic quiz show Jeopardy! In 2011, the question-answering computer system defeated the show's all-time (human) champion, Ken Jennings.

2009

Rajat Raina, Anand Madhavan and Andrew Ng published "Large-Scale Deep Unsupervised Learning Using Graphics Processors," presenting the idea of using GPUs to train large neural networks.

2011

Jürgen Schmidhuber, Dan Claudiu Cire'an, Ueli Meier and Jonathan Masci developed the first CNN to achieve "superhuman" performance by winning the German Traffic Sign Recognition competition.

Apple released Siri, a voice-powered personal assistant that can generate responses and take actions in response to voice requests.

2012

Geoffrey Hinton, Ilya Sutskever and Alex Krizhevsky introduced a deep CNN architecture that won the ImageNet challenge and triggered the explosion of deep learning research and implementation.

2013

China's Tianhe-2 doubled the world's top supercomputing speed at 33.86 petaflops, retaining the title of the world's fastest system for the third consecutive time.

DeepMind introduced deep reinforcement learning, a CNN that learned based on rewards and learned to play games through repetition, surpassing human expert levels.

Google researcher Tomas Mikolov and colleagues introduced Word2vec to automatically identify semantic relationships between words.

2014

Ian Goodfellow and colleagues invented generative adversarial networks, a class of machine learning frameworks used to generate photos, transform images and create deepfakes.

Diederik Kingma and Max Welling introduced variational autoencoders to generate images, videos and text.

Facebook developed the deep learning facial recognition system DeepFace, which identifies human faces in digital images with near-human accuracy.

2016

DeepMind's AlphaGo defeated top Go player Lee Sedol in Seoul, South Korea, drawing comparisons to the Kasparov chess match with Deep Blue nearly 20 years earlier.

Uber started a self-driving car pilot program in Pittsburgh for a select group of users.

2017

Stanford researchers published work on diffusion models in the paper "Deep Unsupervised Learning Using Nonequilibrium Thermodynamics." The technique provides a way to reverse-engineer the process of adding noise to a final image. Google researchers developed the concept of transformers in the seminal paper "Attention Is All You Need," inspiring subsequent research into tools that could automatically parse unlabeled text into large language models (LLMs).

British physicist Stephen Hawking warned, "Unless we learn how to prepare for, and avoid, the potential risks, AI could be the worst event in the history of our civilization."

2018

Developed by IBM, Airbus and the German Aerospace Center DLR, Cimon was the first robot sent into space to assist astronauts. OpenAI released GPT (Generative Pre-trained Transformer), paving the way for subsequent LLMs. Groove X unveiled a home mini-

robot called Lovot that could sense and affect mood changes in humans.

2019

Microsoft launched the Turing Natural Language Generation generative language model with 17 billion parameters. Google AI and Langone Medical Center's deep learning algorithm outperformed radiologists in detecting potential lung cancers.

2020

The University of Oxford developed an AI test called Curial to rapidly identify COVID-19 in emergency room patients. Open AI released the GPT-3 LLM consisting of 175 billion parameters to generate humanlike text models. Nvidia announced the beta version of its Omniverse platform to create 3D models in the physical world. DeepMind's AlphaFold system won the Critical Assessment of Protein Structure Prediction protein-folding contest.

2021

Open AI introduced the Dall-E multimodal AI system that can generate images from text prompts. The University of California, San Diego, created a four-legged soft robot that functioned on pressurized air instead of electronics.

2022

Google software engineer Blake Lemoine was fired for revealing secrets of Lamda and claiming it was sentient. DeepMind unveiled AlphaTensor "for discovering novel, efficient and provably correct algorithms." Intel claimed its FakeCatcher real-time deepfake detector was 96% accurate. OpenAI released ChatGPT in November to provide a chat-based interface to its GPT-3.5 LLM.

2023

OpenAI announced the GPT-4 multimodal LLM that receives both text and image prompts. Elon Musk, Steve Wozniak and thousands more signatories urged a six-month pause on training "AI systems more powerful than GPT-4."

Beyond 2023

We can only begin to envision AI's continuing technological advancements and influences in business processes, manufacturing, healthcare, financial services, marketing, customer experience, workforce environments, education, agriculture, law, IT systems and management, cybersecurity, and ground, air and space transportation.

In business, 55% of organizations that have deployed AI always consider AI for every new use case they're evaluating, according to a 2023 Gartner survey. By 2026, Gartner reported, organizations that "operationalize AI transparency, trust and security will see their AI models achieve a 50% improvement in terms of adoption, business goals and user acceptance."

Today's tangible developments — some incremental, some disruptive — are advancing AI's ultimate goal of achieving artificial general intelligence. Along these lines, neuromorphic processing shows promise in mimicking human brain cells, enabling computer programs to work simultaneously instead of sequentially. Amid these and other mind-boggling advancements, issues of trust, privacy, transparency, accountability, ethics and humanity have emerged and will continue to clash and seek levels of acceptability among business and society.

1.2 Definition of Artificial Intelligence :

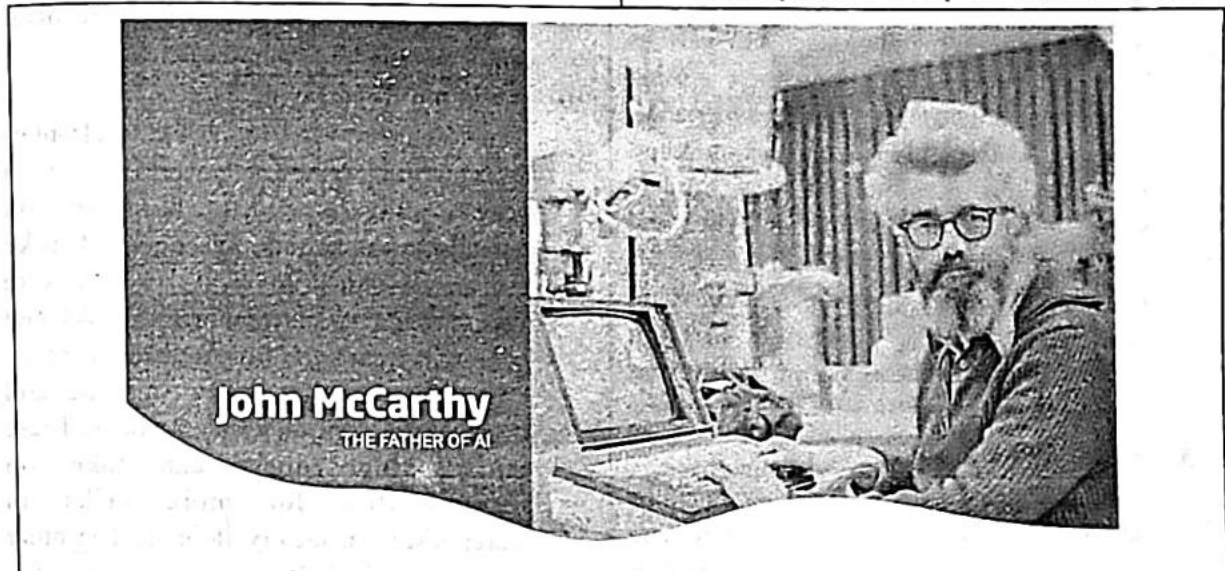
Definition: Artificial Intelligence is the study

of how to make computers do things, which, at the moment, people do better.

According to the father of Artificial Intelligence, John McCarthy, it is "The

science and engineering of making intelligent machines, especially intelligent computer programs".

Artificial Intelligence is a way of making a computer, a computer-controlled robot,



or a software think intelligently, in the similar manner the intelligent humans think.

AI is accomplished by studying how human brain thinks and how humans learn, decide, and work while trying to solve a problem, and then using the outcomes of this study as a basis of developing intelligent software and systems. It has gained prominence recently due, in part, to big data, or the increase in speed, size and variety of data businesses are now collecting.

AI can perform tasks such as identifying patterns in the data more efficiently than humans, enabling businesses to gain more insight out of their data.

From a business perspective AI is a set of very powerful tools, and methodologies for using those tools to solve business problems.

From a programming perspective, AI includes the study of symbolic programming, problem solving, and search.

Applications of Artificial Intelligence:

1. AI in healthcare
2. AI in business
3. AI in education
4. AI in finance and banking
5. AI in security
6. AI in manufacturing
7. AI in software development and IT
8. AI in entertainment and media

1. AI in healthcare :

AI is applied to a range of tasks in the healthcare domain, with the overarching goals of improving patient outcomes and reducing systemic costs. One major application is the use of machine learning models trained on large medical data sets to assist healthcare professionals in making

better and faster diagnoses. For example, AI-powered software can analyze CT scans and alert neurologists to suspected strokes.

2. AI in business :

AI is increasingly integrated into various business functions and industries, aiming to improve efficiency, customer experience, strategic planning and decision-making. For example, machine learning models power many of today's data analytics and customer relationship management (CRM) platforms, helping companies understand how to best serve customers through personalizing offerings and delivering better-tailored marketing.

3. AI in education :

AI has a number of potential applications in education technology. It can automate aspects of grading processes, giving educators more time for other tasks. AI tools can also assess students' performance and adapt to their individual needs, facilitating more personalized learning experiences that enable students to work at their own pace. AI tutors could also provide additional support to students, ensuring they stay on track.

4. AI in finance and banking :

Banks and other financial organizations use AI to improve their decision-making for tasks such as granting loans, setting credit limits and identifying investment opportunities. In addition, algorithmic trading powered by advanced AI and machine learning has transformed financial markets, executing trades at speeds and efficiencies far surpassing what human traders could do manually.

5. AI in security :

AI and machine learning are prominent

buzzwords in security vendor marketing, so buyers should take a cautious approach. Still, AI is indeed a useful technology in multiple aspects of cybersecurity, including anomaly detection, reducing false positives and conducting behavioral threat analytics.

6. AI in manufacturing :

Manufacturing has been at the forefront of incorporating robots into workflows, with recent advancements focusing on collaborative robots, or cobots. Unlike traditional industrial robots, which were programmed to perform single tasks and operated separately from human workers, cobots are smaller, more versatile and designed to work alongside humans. These multitasking robots can take on responsibility for more tasks in warehouses, on factory floors and in other workspaces, including assembly, packaging and quality control.

7. AI in software development and IT :

AI is used to automate many processes in software development, DevOps and IT. For example, AIOps tools enable predictive maintenance of IT environments by analyzing system data to forecast potential issues before they occur, and AI-powered monitoring tools can help flag potential anomalies in real time based on historical system data.

8. AI in entertainment and media :

The entertainment and media business uses AI techniques in targeted advertising, content recommendations, distribution and fraud detection. The technology enables companies to personalize audience members' experiences and optimize delivery of content.

AI technologies are categorised by their

capacity to mimic human characteristics, the technology they use to do this, their real-world applications, and the theory of mind, which we'll discuss in more depth below.

Using these characteristics for reference, all artificial intelligence systems - real and hypothetical - fall into one of three types:

- Artificial Narrow Intelligence (ANI), which has a narrow range of abilities;
- Artificial General Intelligence (AGI), which is on par with human capabilities; or
- Artificial Super Intelligence (ASI), which is more capable than a human.

1.3 Artificial Narrow Intelligence (Weak AI)

Artificial narrow intelligence (ANI), also referred to as weak AI or narrow AI, is the only type of artificial intelligence we have successfully realized to date. Narrow AI is goal-oriented, designed to perform singular tasks - i.e. facial recognition, speech recognition/voice assistants, driving a car, or searching the internet - and is very intelligent at completing the specific task it is programmed to do.

While these machines may seem intelligent, they operate under a narrow set of constraints and limitations, which is why this type is commonly referred to as weak AI. Narrow AI doesn't mimic or replicate human intelligence, it merely simulates human behaviour based on a narrow range of parameters and contexts.

Consider the speech and language recognition of the Siri virtual assistant on iPhones, vision recognition of self-driving cars, and recommendation engines that suggest products you make like based on your purchase history. These systems can

only learn or be taught to complete specific tasks.

Narrow AI has experienced numerous breakthroughs in the last decade, powered by achievements in machine learning and deep learning. For example, AI systems today are used in medicine to diagnose cancer and other diseases with extreme accuracy through replication of human-esque cognition and reasoning.

Narrow AI's machine intelligence comes from the use of natural language processing (NLP) to perform tasks. NLP is evident in chatbots and similar AI technologies. By understanding speech and text in natural language, AI is programmed to interact with humans in a natural, personalised manner.

Narrow AI can either be reactive, or have a limited memory. Reactive AI is incredibly basic; it has no memory or data storage capabilities, emulating the human mind's ability to respond to different kinds of stimuli without prior experience. Limited memory AI is more advanced, equipped with data storage and learning capabilities that enable machines to use historical data to inform decisions.

Most AI is limited memory AI, where machines use large volumes of data for deep learning. Deep learning enables personalised AI experiences, for example, virtual assistants or search engines that store your data and personalise your future experiences.

• EXAMPLES OF NARROW AI :

- Rankbrain by Google / Google Search
- Siri by Apple, Alexa by Amazon, Cortana by Microsoft and other virtual assistants
- IBM's Watson

- Image / facial recognition software
- Disease mapping and prediction tools
- Manufacturing and drone robots
- Email spam filters/social media monitoring tools for dangerous content
- Entertainment or marketing content recommendations based on watch/listen/purchase behaviour
- Self-driving cars
- chatGPT

1.4 Artificial General Intelligence (Strong AI / Deep AI)

Artificial general intelligence (AGI), also referred to as strong AI or deep AI, is the concept of a machine with general intelligence that mimics human intelligence and/or behaviours, with the ability to learn and apply its intelligence to solve any problem. AGI can think, understand, and act in a way that is indistinguishable from that of a human in any given situation.

AGI refers to AI that is designed to perform any intellectual task that a human can do. It is a theoretical form of AI that is not yet possible to achieve. General AI would be able to reason, learn, and understand complex concepts, just like humans. The goal of General AI is to create a machine that can think and learn in the same way that humans do. It would be capable of understanding language, solving problems, making decisions, and even exhibiting emotions. General AI would be able to perform any intellectual task that a human can do, including tasks that it has not been specifically trained to do.

However, General AI also raises some concerns. The development of General AI could have significant ethical implications, as it could potentially surpass human intelligence and become a threat to

humanity. It could also lead to widespread unemployment, as machines would be able to perform tasks that were previously done by humans.

1.5 Artificial Super Intelligence (Hypothetical AI) :

Artificial super intelligence (ASI), is the hypothetical AI that doesn't just mimic or understand human intelligence and behaviour; ASI is where machines become self-aware and surpass the capacity of human intelligence and ability.

Super AI refers to AI that is capable of surpassing human intelligence in all areas. It is a hypothetical form of AI that is not yet possible to achieve. Super AI would be capable of solving complex problems that are beyond human capabilities and would be able to learn and adapt at a rate that far exceeds human intelligence.

The development of Super AI is the ultimate goal of AI research. It would have the ability to perform any task that a human can do, and more. It could potentially solve some of the world's most pressing problems, such as climate change, disease, and poverty.

Possible examples from movies: Skynet (Terminator), Viki (iRobot), Jarvis (Ironman).

The potential of having such powerful machines at our disposal may seem appealing, but the concept itself has a multitude of unknown consequences. If self-aware super intelligent beings came to be, they would be capable of ideas like self-preservation. The impact this will have on humanity, our survival, and our way of life, is pure speculation.

Strong AI	Weak AI
• It is a hypothetical machine that resemble humans in terms of cognitive ability • These machines have a mind of their own which has the ability to think and accomplish complex tasks on their own • Contains common, creativity and expresses emotions • Correlates questions or assignments to a specific data set to accomplish a task • Their goal is to develop creative solutions for each task or issue • There are not many examples of a Strong AI as it is still a hypothetical system under progress in many industries that have gotten close to building a Strong AI	• It is a machine through which intelligent behaviour can be represented and applied by computers to solve complex problems. • These systems are used to accomplish a wide range of problems but operate within a predefined range of functions • No self awareness consciousness or the ability to think • Has the ability to learn and apply knowledge to plan for the future • Their goal is solve particular tasks and problems at a faster pace • Some examples of Weak AI include Amazon's suggested purchases, Alexa and Siri

1.6 Concepts of Production System, Agents and Environments

1.6.1 Production System

AI Production Systems are the backbone of decision-making.

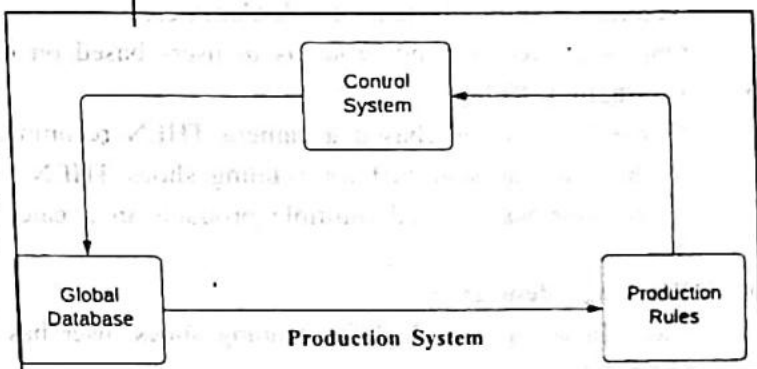
A production system, also known as a rule-based system, is a type of artificial intelligence software designed to mimic the problem-solving ability of human experts. It consists of a knowledge base of rules and an inference engine that applies those rules to solve problems or make decisions within a specific domain.

These systems automate complex tasks through production rules, efficiently processing data and generating insights. They facilitate knowledge-intensive processes comprising a global database, production rules, and a control system. Their key features include simplicity, modularity,

adaptability, and modifiability. Production Systems in AI are classified into various types based on their characteristics, guiding reasoning with control strategies like forward and backward chaining. Understanding production system in Artificial Intelligence and its set of rules is crucial for leveraging AI's potential, integrating them with machine learning, and addressing ethical considerations in their deployment.

The fundamental components of a production system are as follow:

1. Global Database / Working Memory: Also



called the global database, this is a temporary storage area that holds facts about the current state of the problem or situation being analyzed by the system.

2. **Production Rules / Knowledge Base:** This is a collection of rules that encode domain-specific knowledge. Rules typically take the form of "IF (condition) THEN (action)". For example, an expert system for medical diagnosis might have a rule like "IF the patient has a fever AND a rash, THEN there is a possibility of measles."
3. **Control System / Inference Engine:** This is the control mechanism that iteratively evaluates the rules from the knowledge base against the contents of the working memory. It determines which rules are applicable and fires (executes) them, updating the working memory with new facts derived from applying the rules.

Now consider the following categories of production systems:

1. A **monotonic production system** is a production system in which the application of a rule never prevents the later application of another rule that could also have been applied at the time the first rule was selected.
2. A **non-monotonic production system** is one in which (1.) is not true.
3. A **partially commutative production system** is a production system with the property that if the application of a particular sequence of rules transforms state P into state Q, then any combination of those rules that is allowable also transforms state P into state Q.
4. A **commutative production system** is a production system that is both monotonic and partially commutative.

	Monotonic	Non-Monotonic
Partially Commutative	Theorem Proving	Robot Navigation
Not Partially Commutative	Chemical Synthesis	Bridge

• **Examples of production systems :**

Recommendation Systems for E-Commerce

Objective: Recommend products to users based on their browsing history and preferences.

• **Production Rules :**

1. IF the user has purchased a camera THEN recommend camera accessories.
2. IF the user has searched for running shoes THEN suggest related athletic gear.
3. IF the user has viewed multiple products in a category THEN show popular items from that category.

• **Working Memory :**

- Facts: [user has searched for running shoes, user has viewed products in the athletic gear category]

- Rule Interpreter
- Game AI for Strategy Games
- Production Rules
- Working Memory
- Rule Interpreter

• Rule Interpreter :

The system applies the second and third rules to suggest relevant products, enhancing the user's shopping experience.
 Explanation : Recommendation systems use production rules to tailor product suggestions based on user behavior, improving personalization and sales.

• Game AI for Strategy Games

Objective : Control non-player characters (NPCs) in strategy games to challenge players.

• Production Rules :

1. IF the player's army is weak THEN focus on building defenses.
2. IF the player's resources are low THEN prioritize resource gathering.
3. IF the enemy's army is large AND the player's army is strong THEN initiate an attack.

• Working Memory:

- Facts: [player's army is weak, enemy's army is large]

• Rule Interpreter:

- The system evaluates the current game state and applies the appropriate rules to make strategic decisions for the NPCs.

Explanation : In strategy games, production systems help create intelligent behaviors for NPCs, making games more challenging and engaging.

1.6.2 Agents and Environment

In AI, the concepts of "agents" and "environment" are fundamental to understanding how intelligent systems interact

with the world and make decisions. Here's a detailed explanation of each concept:

1. Agents :

Definition : An agent is any entity that perceives its environment through sensors and acts upon that environment through actuators. In AI, agents can range from simple programs to complex robots or autonomous systems.

• Components :

- Sensors : Devices or components that collect data from the environment. For example, a robot may use cameras or microphones as sensors to gather information about its surroundings.
- Actuators : Mechanisms that perform actions based on the agent's decisions. For instance, a robot's actuators might include motors that control its movement or arms that manipulate objects.
- Agent Function : A mapping from the agent's percepts (sensors' inputs) to its actions. This can be described by a policy, which dictates how the agent should act in various situations.
- Agent Program : The software or algorithm that implements the agent function. It processes input from sensors, decides on actions, and uses actuators to perform those actions.

• Types of Agents :

- Simple Reflex Agents : These agents act based on the current percept without considering the history. They follow predefined rules or conditions.
- Model-Based Reflex Agents : They maintain an internal model of the world to handle partial observability and make decisions based on both current percepts and historical data.
- Goal-Based Agents : These agents take

actions based on achieving specific goals. They plan and consider future consequences.

- **Utility-Based Agents :** They act to maximize a utility function, which represents the satisfaction or value of different outcomes.
- **Learning Agents :** These agents improve their performance over time by learning from experiences.

- **Examples :**

- **Virtual Assistants (e.g., Siri, Alexa):** Perceive user commands through voice recognition and act by performing tasks or providing information.
- **Autonomous Vehicles:** Use sensors to perceive the environment (e.g., cameras, radar) and actuators to control steering, acceleration, and braking.

2. Environment :

Definition : The environment is everything that is external to the agent. It encompasses all entities and conditions that the agent interacts with or is affected by. The environment provides the context within which the agent operates.

- **Components :**

- **State :** The specific configuration of the environment at a given time. The state can be dynamic and change over time based on the actions of the agent and other factors.
- **Percepts :** The information or data received by the agent from the environment through its sensors. Percepts inform the agent about the current state of the environment.

- **Actions :** The operations or behaviors that the agent can perform that affect the environment. Actions lead to changes in the state of the environment.

- **Characteristics :**

- **Observable vs. Partially Observable :** In a fully observable environment, the agent can access all information about the current state. In a partially observable environment, the agent only receives limited or incomplete information.
- **Static vs. Dynamic :** A static environment remains unchanged while the agent is deliberating. A dynamic environment changes over time, which can affect the agent's decisions.
- **Discrete vs. Continuous :** In a discrete environment, the states and actions are distinct and countable. In a continuous environment, there are an infinite number of possible states and actions.

- **Examples :**

- **Game Environments:** In a game like chess, the environment consists of the chessboard, pieces, and the rules of the game. The state of the environment changes as players make moves.
- **Real-World Environments:** For autonomous vehicles, the environment includes roads, traffic signals, other vehicles, and pedestrians.

Interaction Between Agents and Environments

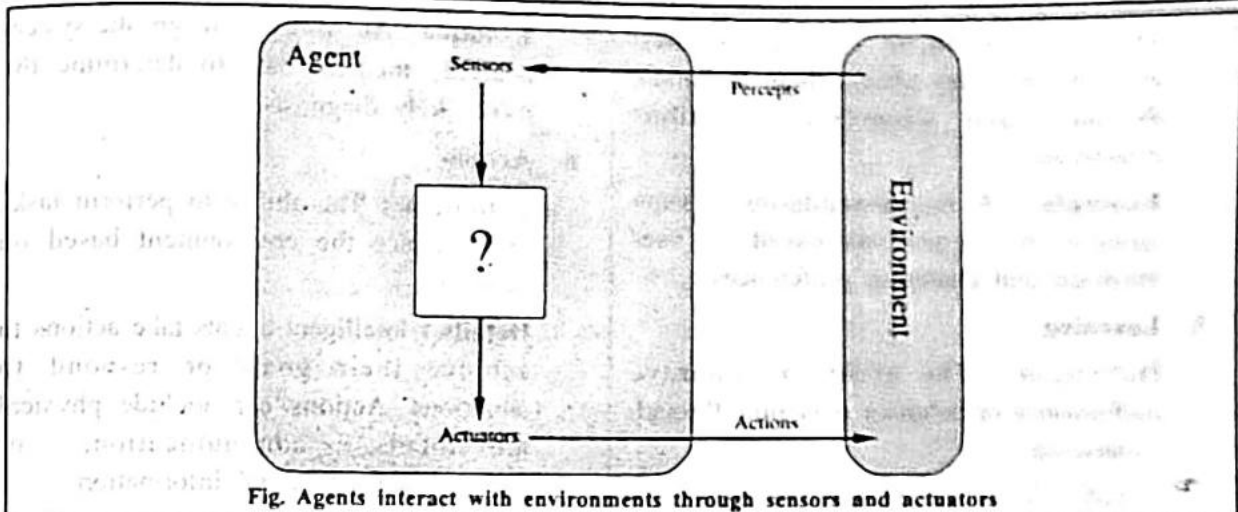


Fig. Agents interact with environments through sensors and actuators

• Perception and Action Cycle :

1. **Perception** : The agent senses the environment and receives data (percepts).
2. **Decision Making** : Based on the percepts, the agent processes this information using its internal model or decision-making algorithm.
3. **Action** : The agent takes actions that affect the environment, potentially leading to a new state.

4. **Feedback Loop** : The environment responds to the agent's actions, leading to new percepts, and the cycle continues.

• Example Scenario :

- **Autonomous Delivery Robot** : The robot (agent) perceives its environment (e.g., obstacles, destinations) using sensors. It makes decisions based on its goal (e.g., delivering a package). It then acts by navigating through the environment. The environment changes as the robot moves, and the process repeats.

1.7 Characteristic of Intelligent Agents :

Intelligent agents possess several key characteristics that enable them to operate effectively in various environments and perform tasks autonomously. Here are the

primary characteristics of intelligent agents:

- | | |
|---------------------------|-----------------|
| 1. Autonomy | 2. Adaptability |
| 3. Learning | 4. Perception |
| 5. Reasoning | 6. Action |
| 7. Goal-Oriented Behavior | 9. Robustness |
| 8. Interactivity | |
| 10. Efficiency | |

1. Autonomy

Definition : The ability to operate independently without human intervention.

Details : Intelligent agents make decisions and perform actions based on their own perceptions and internal logic rather than relying on external commands. They can adapt to changes in the environment and modify their behavior accordingly.

Example : An autonomous vehicle navigates and makes driving decisions without direct input from a human driver.

2. Adaptability

Definition : The capability to adjust their behavior based on changes in the environment or internal state.

Details : Intelligent agents can modify their strategies and actions in response to

new information or changes in their environment. This allows them to handle dynamic and uncertain conditions effectively.

Example : A recommendation system updates its suggestions based on user feedback and changing preferences.

3. Learning

Definition : The ability to improve performance or behavior over time through experience.

Details : Intelligent agents can learn from past experiences and interactions with the environment to enhance their performance. This includes recognizing patterns, refining strategies, and adapting to new situations.

Example : A chatbot improves its responses by learning from previous interactions with users.

4. Perception

Definition : The process of acquiring and interpreting data from the environment.

Details : Intelligent agents use sensors or input mechanisms to gather information about their surroundings. This data is used to understand the current state of the environment and inform decision-making.

Example : A robot uses cameras and sensors to detect obstacles and navigate its environment.

5. Reasoning

Definition : The ability to process information and make decisions based on logical analysis.

Details : Intelligent agents use reasoning to interpret data, evaluate options, and make informed decisions. This can involve deductive reasoning, inductive reasoning, or probabilistic reasoning.

Example : An AI-driven diagnostic system analyzes medical data to determine the most likely diagnosis.

6. Action

Definition : The ability to perform tasks or influence the environment based on decisions.

Details : Intelligent agents take actions to achieve their goals or respond to situations. Actions can include physical movements, communication, or manipulation of digital information.

Example : An industrial robot performs assembly tasks or adjusts machine settings based on its programmed instructions.

7. Goal-Oriented Behavior :

Definition : The pursuit of specific objectives or goals.

Details : Intelligent agents operate with goals in mind, which guide their actions and decision-making processes. They evaluate different strategies to achieve these goals effectively.

Example : A personal assistant AI schedules appointments and manages tasks to help users achieve their productivity goals.

8. Interactivity

Definition : The ability to interact with other agents or humans.

Details : Intelligent agents often need to communicate and coordinate with other agents or users. This involves understanding and responding to input from others and working collaboratively when necessary.

Example : A multi-agent system for traffic management coordinates between different traffic lights and sensors to optimize traffic flow.

9. Robustness

Definition : The ability to handle errors, uncertainties, and unexpected situations.

Details : Intelligent agents are designed to operate reliably even in the face of incomplete information, uncertainties, or unexpected changes. They can manage errors and adapt their strategies as needed.

Example : A drone adjusts its flight path in response to sudden weather changes or obstacles.

10. Efficiency

Definition : The ability to achieve goals with optimal use of resources.

Details : Intelligent agents strive to perform tasks and make decisions in a way that maximizes efficiency, minimizing resource usage such as time, energy, and computational power.

Example : An optimization algorithm for scheduling tasks aims to minimize the total time and cost of completing all tasks.

1.8 Concept of Rationality :

Rationality in AI refers to the principle that an intelligent agent should act in a manner that maximizes its chances of achieving its goals, given its knowledge and resources. An agent is deemed rational if it consistently makes decisions and performs actions that optimize its performance measure or utility, reflecting its effectiveness in reaching its objectives.

What is rational at any given time depends on four things :

1. The performance measure that defines the criterion of success.
2. The agent's prior knowledge of the environment.
3. The actions that the agent can perform.
4. The agent's percept sequence to date.

This leads to a definition of a rational

agent: For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

• Key Aspects of Rationality :

1. **Rational Agent :** A rational agent is one that selects actions that are expected to maximize its performance measure based on its current knowledge and understanding of the environment. It acts to achieve specific goals or maximize utility, which is a quantitative measure of satisfaction or preference.
2. **Performance Measure :** This is a metric used to assess how well an agent is achieving its goals. For instance, in a game, the performance measure might be winning or scoring points. In an autonomous vehicle, it might be safety, efficiency, and adherence to traffic rules.
3. **Utility Function :** Rational agents often use a utility function to evaluate the desirability of different outcomes. The goal is to choose actions that maximize expected utility, reflecting the agent's preferences and goals.
4. **Bounded Rationality :** In practice, agents often face limitations such as incomplete information, limited computational resources, and time constraints. Bounded rationality acknowledges these constraints and suggests that agents may use heuristics or rules of thumb to make decisions that are satisfactory rather than optimal.
5. **Decision-Making Process :** A rational decision-making process involves perceiving the environment, evaluating possible actions based on the performance

measure or utility function, making a decision, and acting accordingly. The agent then observes the outcomes to update its knowledge and refine future decisions.

Rationality in AI aims to create systems that behave predictably and effectively in various scenarios, optimizing their performance despite practical limitations. This concept is crucial for designing intelligent systems that can operate autonomously and adapt to changing environments.

In our discussion of the rationality of the simple vacuum-cleaner agent, we had to specify the performance measure, the environment, and the agent's actuators and sensors. We group all these under the heading of the task environment. For the acronymically minded, we call this the PEAS (Performance, Environment, Actuators, Sensors) description. In designing an agent, the first step must always be to specify the task environment as fully as possible.

Examples of Agent types with their PEAS

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display of scene categorization	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections	Keyboard entry

1.9 Nature of Environments :

In AI, the environment encompasses everything external to the intelligent agent that influences its behaviour and decision-making. The nature of the environment significantly impacts how an agent operates and interacts with the world. Understanding the characteristics of the environment is crucial for designing effective and adaptive intelligent systems.

• **Key Aspects of the Environment :**

1. Observability
2. Dynamics
3. Discrete vs Continuous
4. Episodic vs. Sequential
5. Deterministic vs. Stochastic
6. Static vs. Changing

1. Observability :

- **Fully Observable :** The agent has access to complete information about the current state of the environment. For example, a chess game is fully observable because all pieces are visible at all times.
- **Partially Observable :** The agent only has access to incomplete information. For instance, in a poker game, players cannot see their opponents' cards, making it partially observable.

2. Dynamics :

- **Static :** The environment does not change while the agent is deliberating. An example is a board game where the environment remains constant until players make moves.
- **Dynamic :** The environment changes over time, often independently of the agent's actions. For example, traffic conditions change continuously, affecting an autonomous vehicle's decisions.
- **Semi-Dynamic :** If the environment itself does not change with the passage of time but the agent's performance score does, then we say the environment is semi-dynamic. Chess, when played with a clock, is semi-dynamic.

3. Discrete vs Continuous :

- **Discrete :** The environment is characterized by distinct, countable states and actions. For instance, a grid-based

game like "Pac-Man" has a discrete environment with a finite number of positions and actions.

- **Continuous :** The environment has an infinite number of states and actions. For example, real-world navigation involves continuous variables like speed and position.

4. Episodic vs. Sequential :

- **Episodic :** Each decision or action is independent of previous ones. The environment's state does not affect future actions. For example, a customer service chatbot responding to isolated inquiries operates in an episodic environment.
- **Sequential :** Actions have a lasting impact, and the current state affects future states and decisions. For instance, a robot navigating a maze must consider the sequence of actions to reach its goal.

5. Deterministic vs. Stochastic :

- **Deterministic :** The outcome of an action is predictable and consistent. For example, in a controlled laboratory environment, actions lead to predictable results.
- **Stochastic :** The outcome of an action involves uncertainty or randomness. For example, weather prediction involves stochastic elements due to the inherent unpredictability of weather patterns.

6. Static vs. Changing :

- **Static :** The environment remains unchanged during the agent's decision-making process. For instance, a puzzle game with a fixed layout is static.
- **Changing :** The environment can change dynamically during the agent's interactions. For example, financial markets change frequently, requiring adaptive decision-making.

Examples of task environments with their characteristics

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Interactive English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

❖ Question Bank ❖

- Define Artificial Intelligence. State applications of AI.
- Discuss types of AI in detail.
- Elaborate history of AI in brief.
- Give the difference between Weak AI and Strong AI.
- Explain production system in AI with a suitable example.
- Explain concept of rationality in AI.
- Discuss any seven characteristics of an intelligent agent.
- Compare and contrast discrete environment with continuous environment.
- Compare and contrast stochastic environment with deterministic environment.
- Explain interaction between agents and environment in AI.
- Is AI a science, or is it engineering? Or neither or both? Explain.
- To what extent are the following computer systems instances of artificial intelligence:
 - Supermarket bar code scanners.
 - Websearch engines.
 - Voice-activated telephone menus.
 - Internet routing algorithms that respond dynamically to the state of the network.
- Define in your own words: (a) intelligence, (b) artificial intelligence, (c) agent, (d) rationality, (e) logical reasoning.
- Are reflex actions (such as flinching from a hot stove) rational? Are they intelligent?
- For each of the following activities, give a PEAS description of the task environment and characterize it in terms of the properties.
 - Playing soccer.
 - Exploring the subsurface oceans of Titan.
 - Shopping for used AI books on the Internet.
 - Playing a tennis match.
 - Practicing tennis against a wall.
 - Performing a high jump.
 - Knitting a sweater.
 - Bidding on an item at an auction.

- State following statements are whether True or False.
1. **True or False :** The term "Artificial Intelligence" was first coined in 1956 at a conference at Dartmouth College.
 2. **True or False :** The development of AI began in the 21st century, following significant advancements in computer technology.
 3. **True or False :** Early AI research primarily focused on symbolic reasoning and problem-solving.
 4. **True or False :** Artificial Intelligence refers only to the ability of machines to perform tasks that require human-like intelligence.
 5. **True or False :** AI systems can be classified into narrow AI, general AI, and super AI based on their capabilities.
 6. **True or False :** Artificial Narrow Intelligence (ANI) is designed to perform a specific task, such as facial recognition.
 7. **True or False :** Artificial General Intelligence (AGI) refers to systems that can perform any intellectual task that a human can do.
 8. **True or False :** Artificial Super Intelligence (ASI) surpasses human intelligence across all domains.
 9. **True or False :** A production system in AI consists of a set of rules that dictate the behaviour of an agent.
 10. **True or False :** An agent in AI is defined as any entity that perceives its environment and takes actions to achieve specific goals.
 11. **True or False :** The environment in which an agent operates does not influence its decision-making process.
 12. **True or False :** Intelligent agents must always act rationally to achieve their goals effectively.
 13. **True or False :** The concept of rationality in AI implies that an agent makes decisions based solely on past experiences.
 14. **True or False :** Environments can be categorized as deterministic or stochastic based on the predictability of outcomes.
 15. **True or False :** An agent that can learn from its experiences is referred to as a learning agent.
 16. **True or False :** The nature of an environment is irrelevant when designing an intelligent agent.

❖ MCQ ❖

1. **What is the primary goal of Artificial Intelligence (AI) ?**
 - a) To create robots that can work autonomously
 - b) To simulate human intelligence in machines
 - c) To enhance data processing capabilities
 - d) To develop new programming languages
 2. **Which of the following best defines Artificial Narrow Intelligence (ANI)?**
 - a) AI that can learn and adapt to new tasks
 - b) AI focused on a single task or narrow range of tasks
 - c) AI that can outperform humans in every field
- Answer:** b) To simulate human intelligence in machines

- d) AI that requires constant human supervision

Answer: b) AI focused on a single task or narrow range of tasks

3. Artificial General Intelligence (AGI) refers to :

- a) AI that performs only specialized tasks
- b) AI with human-like cognitive abilities across various domains
- c) AI designed for high-speed calculations
- d) AI that operates without any data input

Answer: b) AI with human-like cognitive abilities across various domains

4. What does Artificial Super Intelligence (ASI) represent ?

- a) AI that is equal to human intelligence
- b) AI that is less capable than humans
- c) AI that exceeds human intelligence in virtually all aspects
- d) AI that can only replicate human emotions

Answer: c) AI that exceeds human intelligence in virtually all aspects

5. In AI terminology, what is a "production"?

- a) The result of an agent's decision-making process
- b) A defined rule that describes how to act in response to certain conditions
- c) The process of developing new AI systems
- d) The collection of data used for training

Answer: b) A defined rule that describes how to act in response to certain conditions

6. Which of the following best describes an "agent" in AI ?

- a) A hardware device that processes data
- b) Any entity that can perceive its environment and take action
- c) A programmer who develops AI systems
- d) A software application used to manage databases

Answer: b) Any entity that can perceive its environment and take action

7. What does "environment" refer to in the context of AI ?

- a) The physical location where AI operates
- b) The context in which an agent functions and makes decisions
- c) The data used to train AI models
- d) The algorithms that drive AI systems

Answer: b) The context in which an agent functions and makes decisions

8. Which of the following is a key characteristic of intelligent agents ?

- a) They require human intervention at all times
- b) They can learn from experience and adapt their behavior
- c) They can only execute predefined tasks
- d) They operate in isolation without external input

Answer: b) They can learn from experience and adapt their behavior

9. The concept of rationality in AI implies:

- a) Making decisions based on emotions
- b) Choosing the best action based on available information
- c) Following rules without deviation

- d) Performing tasks with minimal human supervision

Answer: b) Choosing the best action based on available information

10. Environments in AI can be categorized into which of the following?

- a) Static and dynamic
- b) Simple and complex
- c) Deterministic and stochastic
- d) All of the above

Answer: d) All of the above

11. The term "machine learning" is often associated with which type of AI ?

- a) Artificial Narrow Intelligence
- b) Artificial General Intelligence
- c) Artificial Super Intelligence
- d) All types of AI

Answer : a) Artificial Narrow Intelligence

12. What is a characteristic of a dynamic environment in AI?

- a) The conditions remain constant over time
- b) The environment can change unpredictably
- c) The agent has complete control over the environment
- d) The environment is predictable and stable

Answer: b) The environment can change unpredictably

13. In which type of AI is self-improvement a key feature ?

- a) Artificial Narrow Intelligence
- b) Artificial General Intelligence
- c) Artificial Super Intelligence
- d) All types of AI

Answer: c) Artificial Super Intelligence

14. What is the main challenge associated with developing Artificial General Intelligence ?

- a) The availability of computational resources
- b) Achieving human-like reasoning and adaptability
- c) The need for large datasets
- d) Ensuring safety protocols are in place

Answer: b) Achieving human-like reasoning and adaptability

15. Which of the following is an example of an intelligent agent ?

- a) A simple calculator
- b) A chess-playing program
- c) A static website
- d) A text document

Answer: b) A chess-playing program

16. In AI, what is a "state"?

- a) The current condition or configuration of the environment
- b) The action taken by an agent
- c) The rules governing agent behavior
- d) The output produced by an agent

Answer : a) The current condition or configuration of the environment

17. Which of the following best describes a stochastic environment ?

- a) The outcomes are predictable and consistent
- b) The outcomes are uncertain and influenced by randomness.
- c) The environment remains unchanged over time
- d) The environment is controlled entirely by the agent

Answer: b) The outcomes are uncertain and influenced by randomness

18. What role does feedback play in the learning process of intelligent agents ?

- a) It is unnecessary for learning
- b) It helps agents refine their actions based on past experiences
- c) It only complicates the learning process
- d) It is used solely for testing purposes

Answer: b) It helps agents refine their actions based on past experiences

19. Which of the following is NOT a feature of Artificial Narrow Intelligence?

- a) Specialization in specific tasks
- b) Ability to generalize across multiple tasks
- c) High efficiency in predefined areas
- d) Limited cognitive capabilities

Answer: b) Ability to generalize across multiple tasks

20. What is the significance of the Turing Test in AI ?

- a) It measures the speed of an AI system
- b) It assesses the ability of a machine to exhibit intelligent behavior indistinguishable from a human
- c) It determines the computational power of an AI
- d) It evaluates the data used in training AI models

Answer: b) It assesses the ability of a machine to exhibit intelligent behavior indistinguishable from a human

1. What is your understanding of Artificial Intelligence (AI)? Can you provide a concise definition?
2. How would you differentiate between

❖ Interview oriented Questions ❖

Artificial Narrow Intelligence (ANI), Artificial General Intelligence (AGI), and Artificial Super Intelligence (ASI)?

3. Can you explain the significance of AI in today's technology landscape? What are some practical applications?
4. What are some examples of systems that currently utilize Artificial Narrow Intelligence?
5. What challenges do researchers face when developing Artificial General Intelligence?
6. How do you envision the future of Artificial Super Intelligence? What implications might it have for society?
7. Can you explain the concept of a "production" in the context of AI? How is it used in decision-making?
8. What is an intelligent agent, and how does

it operate within its environment?

9. How would you define the "environment" in which an AI agent operates? What factors influence this environment?
10. What are the key characteristics of intelligent agents? How do these characteristics affect their performance?
11. How does an intelligent agent learn from its experiences? Can you give an example?
12. What does the concept of rationality mean in the context of AI? How is it applied in decision-making processes?
13. How do you differentiate between static and dynamic environments in AI? Can you provide examples of each?
14. What is the difference between deterministic and stochastic environments? How

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| <p>does this impact an agent's decision-making process?</p> <p>15. Can you discuss how an intelligent agent might adapt to a changing environment? What strategies might it employ?</p> <p>16. Can you give an example of how AI is used in a real-world application? What type of intelligence does it utilize?</p> <p>17. What ethical considerations do you think are important when developing AI systems, particularly in relation to AGI and ASI?</p> | <p>18. How do you see the role of human oversight in the development and deployment of AI technologies?</p> <p>19. What trends do you anticipate in the field of AI over the next five to ten years?</p> <p>20. In your opinion, what is the most exciting aspect of working in the field of AI today? Why?</p> |
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